Aim

To implement an **Autoencoder using TensorFlow/Keras** to remove noise from images in the MNIST dataset.

❓ Problem Statement

Images often contain noise which reduces quality and affects analysis.
The problem is to reconstruct clean images from noisy input images using an autoencoder model.

📊 Dataset Link

https://www.kaggle.com/datasets/oddrationale/mnist-in-csv

⚙️ Methodology

- Loaded MNIST dataset using Pandas
- Separated pixel values and normalized data (0–255 → 0–1)
- Added random noise to images
- Split data into training and testing sets
- Built autoencoder with:
    - Encoder (Dense layers)
    - Decoder (Dense layers)
- Compiled model using Adam optimizer and MSE loss
- Trained model for multiple epochs
- Generated denoised images from noisy inputs

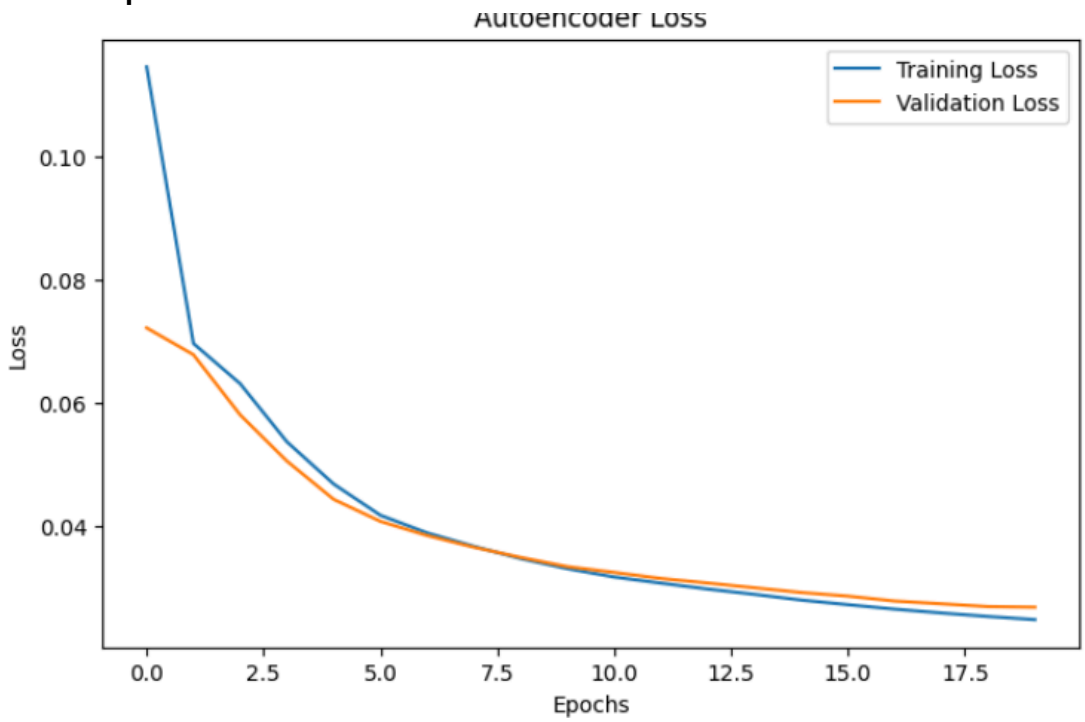
📈 Result

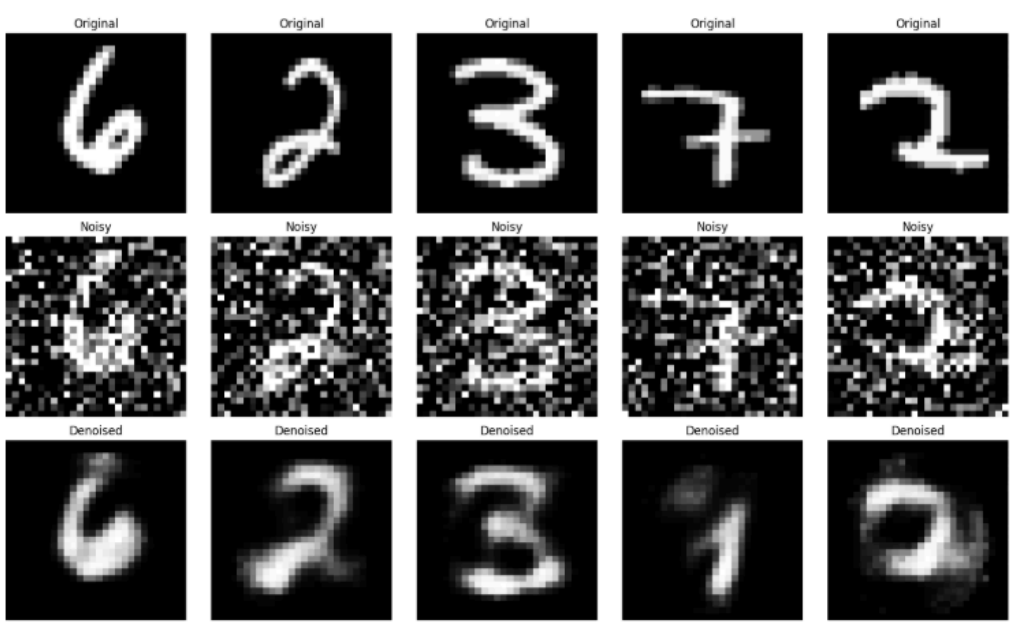
- **Loss (MSE):** Decreased over epochs
- Autoencoder successfully reconstructed clean images

Performance Metrics:

- Mean Squared Error (MSE)

Model Graphs:





🧾 Conclusion

- Autoencoder effectively removed noise from images.
- Model learned meaningful image representations.
- Useful for image restoration tasks.